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IEEE Signal Processing Society

# SPCUP 2025

## Team-IITH

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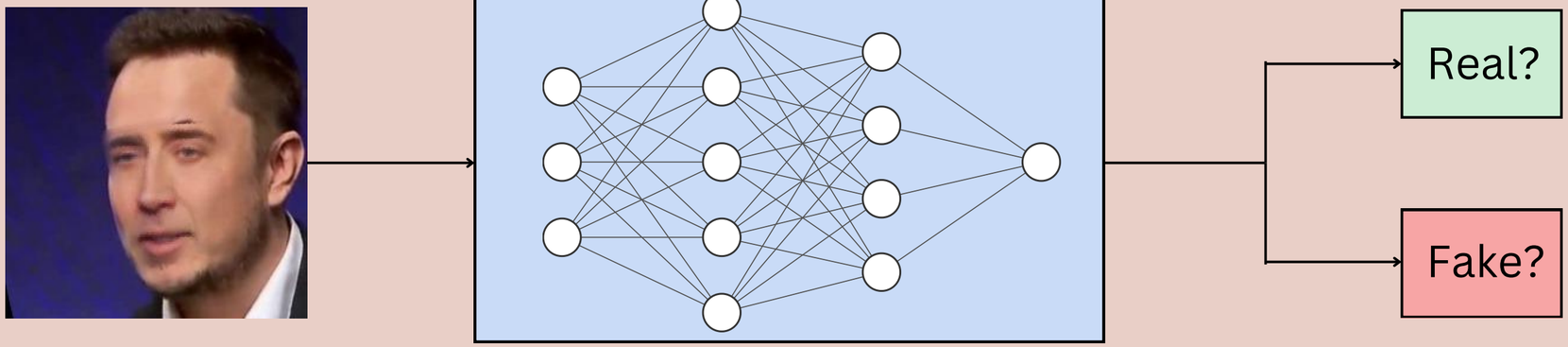
# Table of Contents


- Problem Statement
- Model Architecture
- Standard Models results
- Models used
  - ConvNeXt
  - Max ViT
  - SWIN Transformer
- Model Ensemble
- Our Results
- Explanation Maps

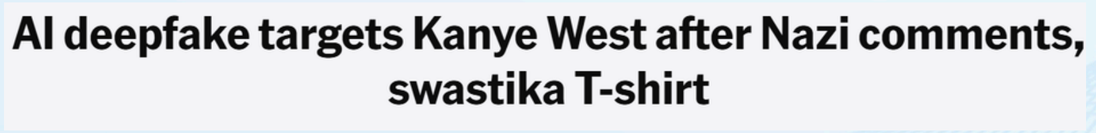
# Problem Statement




## Dataset Details :

- Training Data:
  - 219K fake images
  - 42K real images
- Validation Data:
  - 1.5K fake images
  - 1.5K real images
- Testing Data:
  - 395.6K images

Speed up plans to criminalise deepfake abuse, ministers told


AI deepfake targets Kanye West after Nazi comments, swastika T-shirt

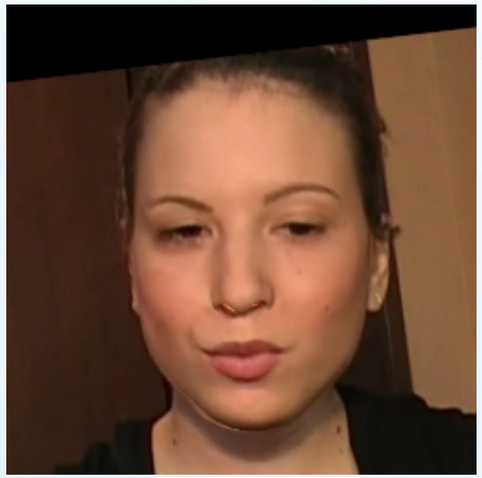
RASHMIKA MANDANNA CONTROVERSY:
HOW YOU CAN SPOT A DEEPFAKE

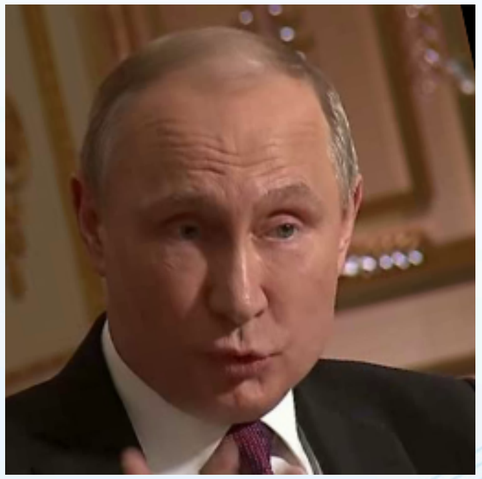
The original video features a British Indian woman (*left*). Mandanna's face is morphed onto it.

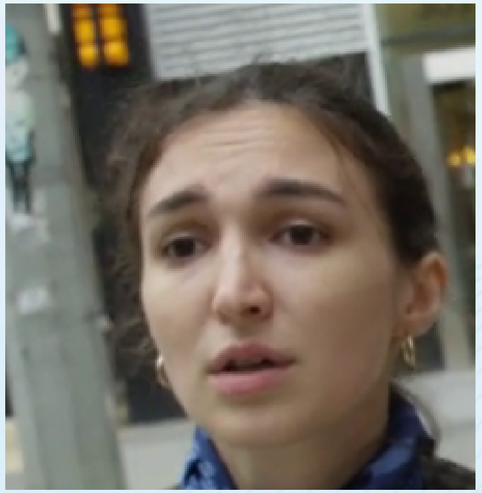
The danger of deepfakes

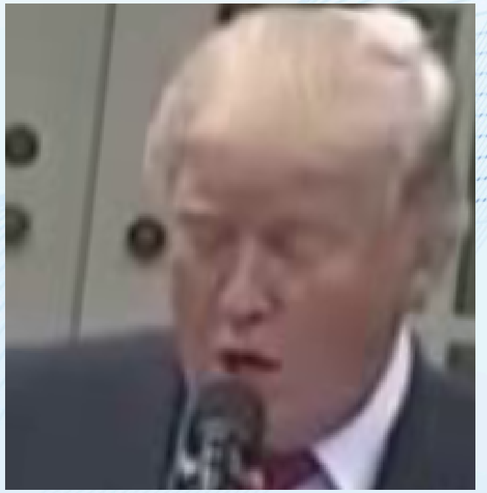
What are the ways in which AI-manipulated digital media can impact the lives of individuals as well as influence the public discourse? How is it employed by various groups and how can society overcome the 'infodemic'?

# How Many are Real ?







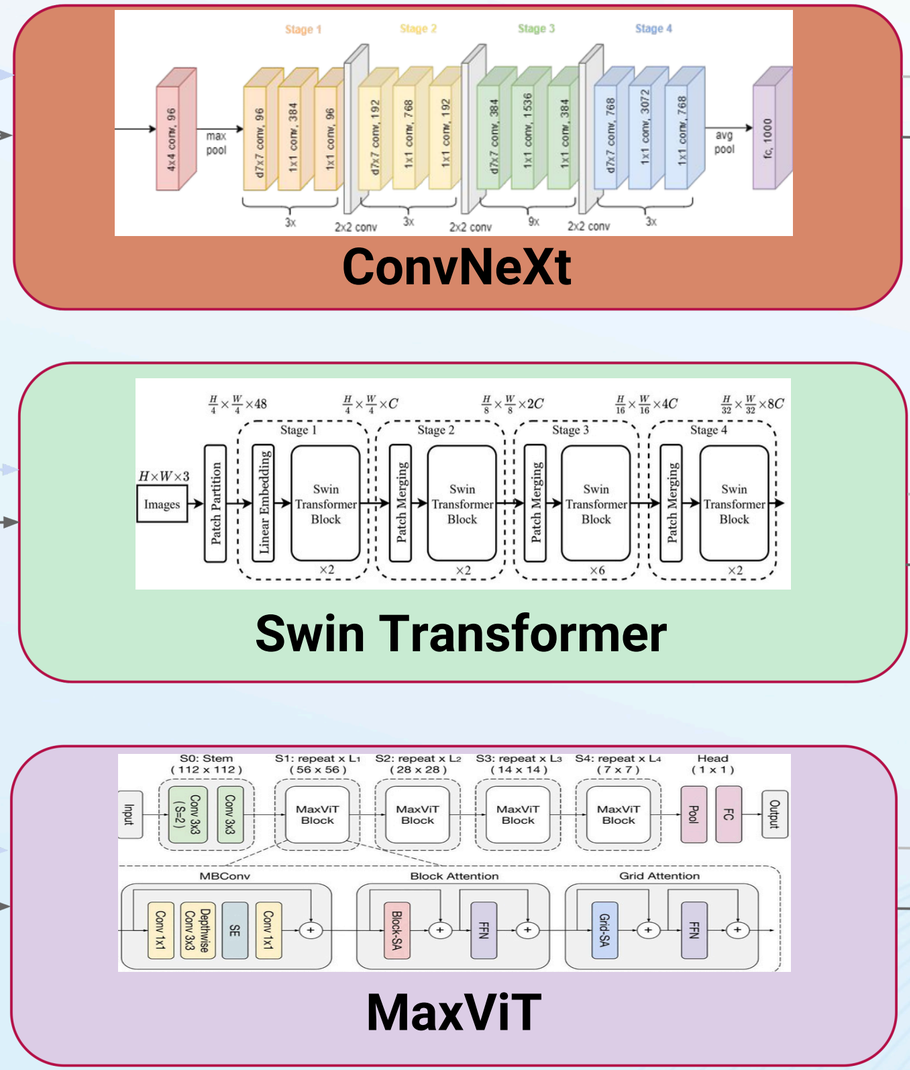


# How Many are Real ?

# Our Model Architecture

Real

Fake

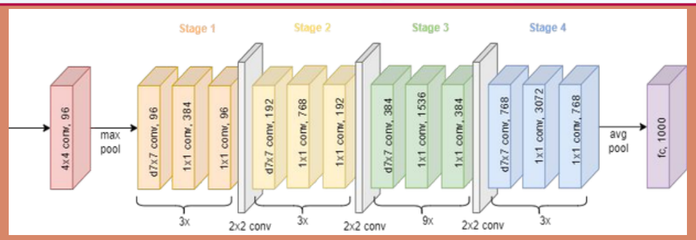



Ensemble 1

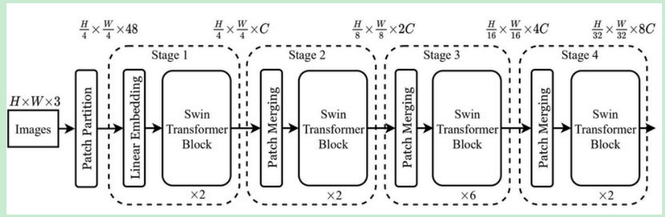
# Our Model Architecture

Real

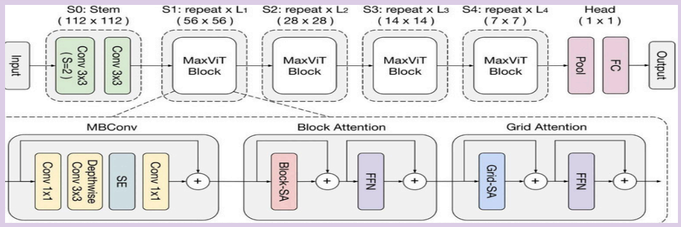
Fake




**ConvNeXt**




**Swin Transformer**




**MaxViT**

**Ensemble 1**

| Confidence Scores | |
|---|---|
| Real | 0.987 |
| Fake | 0.017 |

# Standard Models

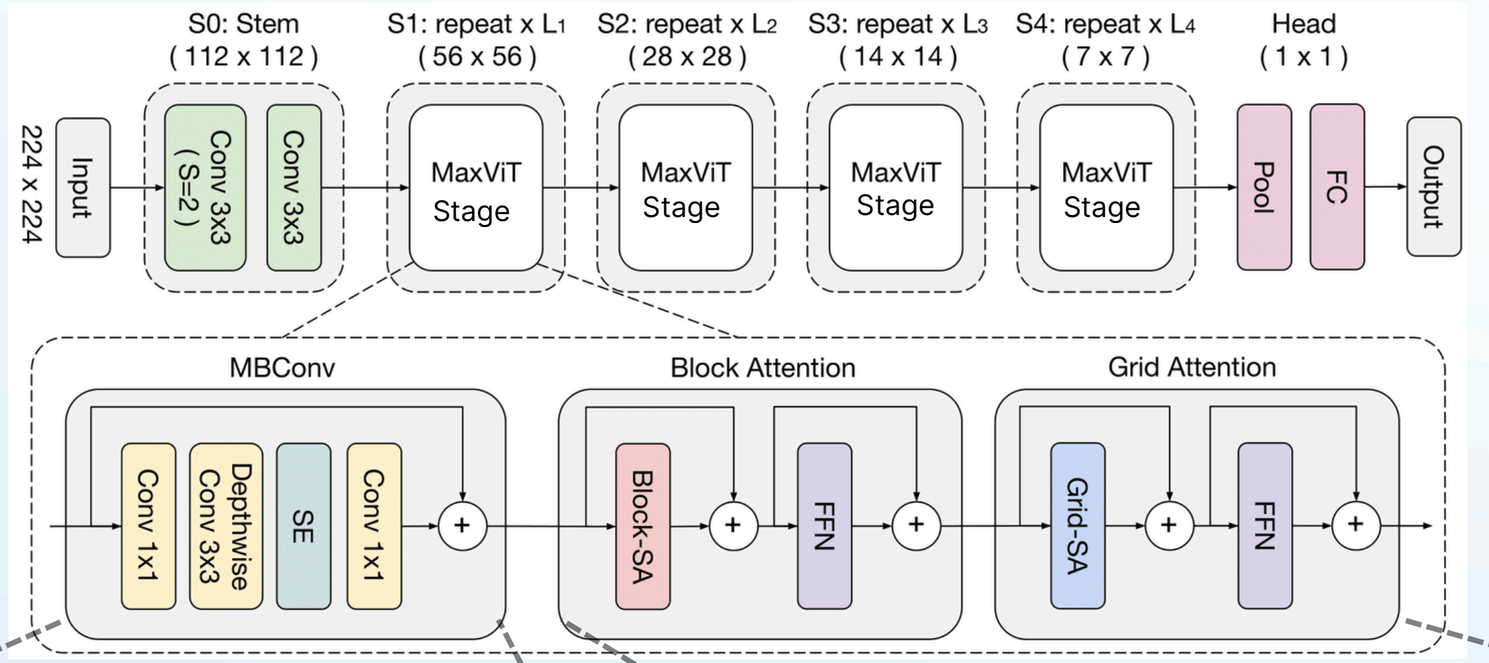
| Model | Accuracy % |
| --- | --- |
| ViT | 93 |
| ResNet-101 | 94.6 |
| EfficientNet-b4 | 91.08 |
| Inception-v4 | 94.63 |

# DeepFakeBench

| Detector | Accuracy % |
|---|---|
| UCF | 92.84 |
| FFD | 90.98 |
| Core | 91.08 |
| F3NET | 92.94 |
| SPSL | 92.51 |
| SRM | 90.17 |

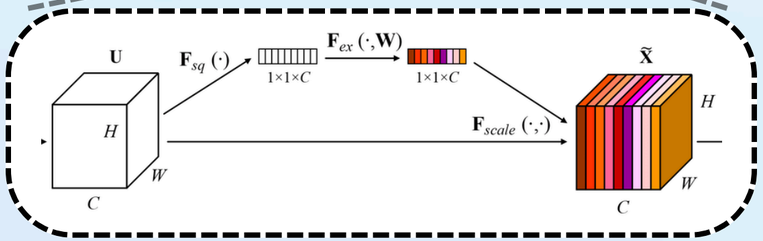
# MaxViT

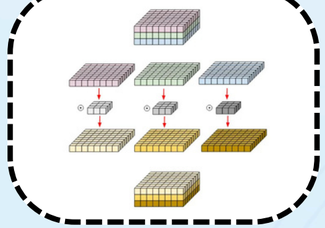
# MaxViT: Multi-Axis Vision Transformer

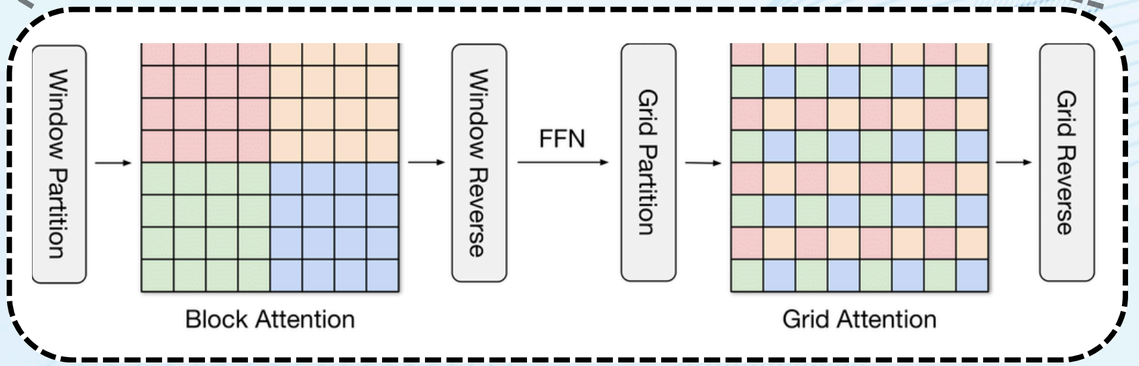



$$\mathbf{x} \leftarrow \text{Proj}(\text{Pool2D}(\mathbf{x})) + \text{Proj}(\text{SE}(\text{DWConv}\downarrow(\text{Conv}(\text{Norm}(\mathbf{x})))))$$

$$\mathbf{x} \leftarrow \mathbf{x} + \text{Proj}(\text{SE}(\text{DWConv}(\text{Conv}(\text{Norm}(\mathbf{x})))))$$






Tu, Z., Talebi, H., Zhang, H., et al. "MaxViT: Multi-Axis Vision Transformer." ECCV, 2022

# Hyperparameters

**Basic**

- Batch size = 64
- Epochs = 20

**Augmentations**

- RandAug = 15
- RandomHorizontalFlip (p=0.25)

**Training tricks**

- Label Smoothing (ϵ = 0.1)
- Cosine LR Schedule (1e-4 →1e-5)

**Optimizer**

- AdamW optimizer
- lr = 1e-4
- $\beta_1$ = 0.9
- $\beta_2$ = 0.999
- weight decay = 0.01

# Ablation Study

| Configuration | Accuracy % |
|---|---|
| Baseline | 93.16 |
| + RandAug (N = 2, M = 15) + RandHFlip (p = 0.25) | 95.57 |
| + Label Smoothing (ϵ = 0.1) | 95.86 |
| + Cosine LR Schedule (1e-4 → 1e-5) | **96.09** |

+ 2.41 %

+ 0.29 %

+ 0.23 %

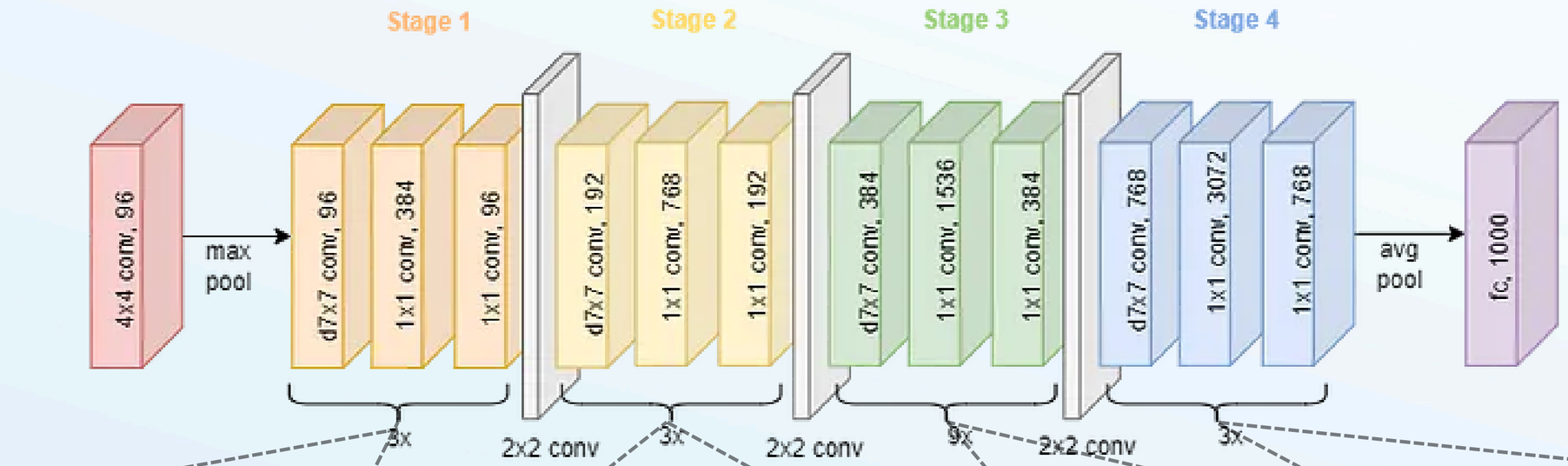
+ Cosine LR scheduling (1e-4 → 5e-5) : 95.57%

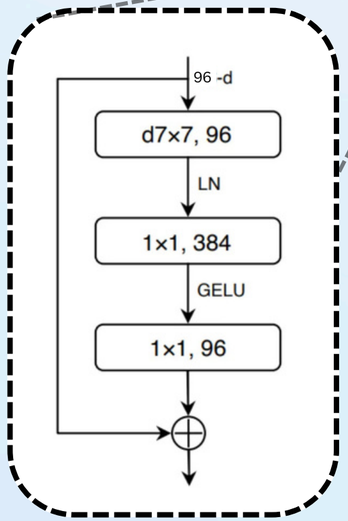
# Results

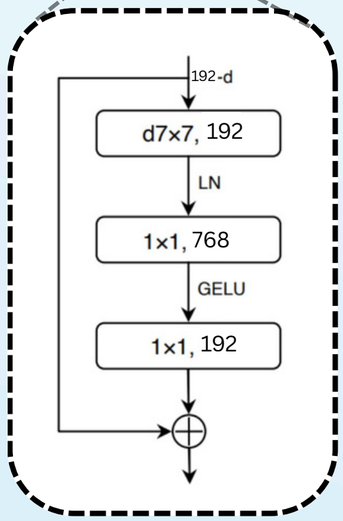
| Model | Accuracy % |
|---|---|
| ViT | 93 |
| ResNet-101 | 94.6 |
| EfficientNet-b4 | 91.08 |
| Inception-v4 | 94.63 |
| **MaxViT** | **96.09** |

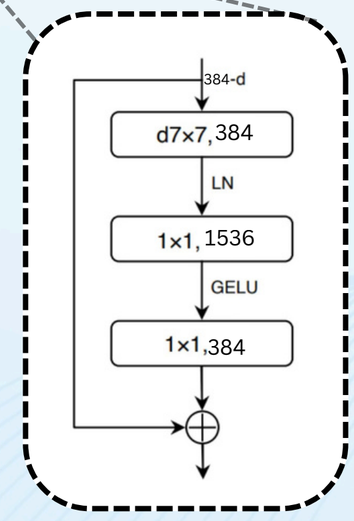
# ConvNeXt

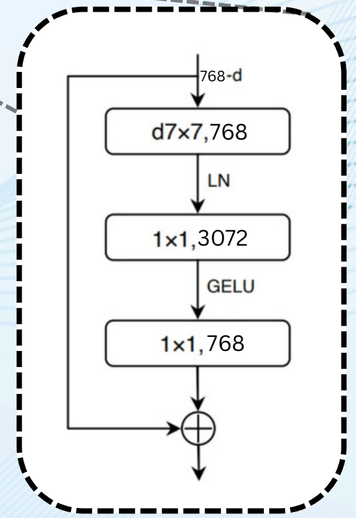
# ConvNeXt




**Stage 1 ConvNeXt Block** **Stage 2 ConvNeXt Block** **Stage 3 ConvNeXt Block** **Stage 4 ConvNeXt Block**

Liu, Z., Mao, H., Wu, C. Y., Feichtenhofer, C., Darrell, T., & Xie, S. (2022). A convnet for the 2020s. CVPR 2022

# Hyperparameters

**Basic**

- Batch size = 32
- Epochs = 10

**Augmentations**

- Rotation: [-10°, +10°]
- Gaussian Blur: 30% probability (Kernel: 3x5)
- Brightness & Contrast: 50% probability
- Horizontal Flip: 50% probability

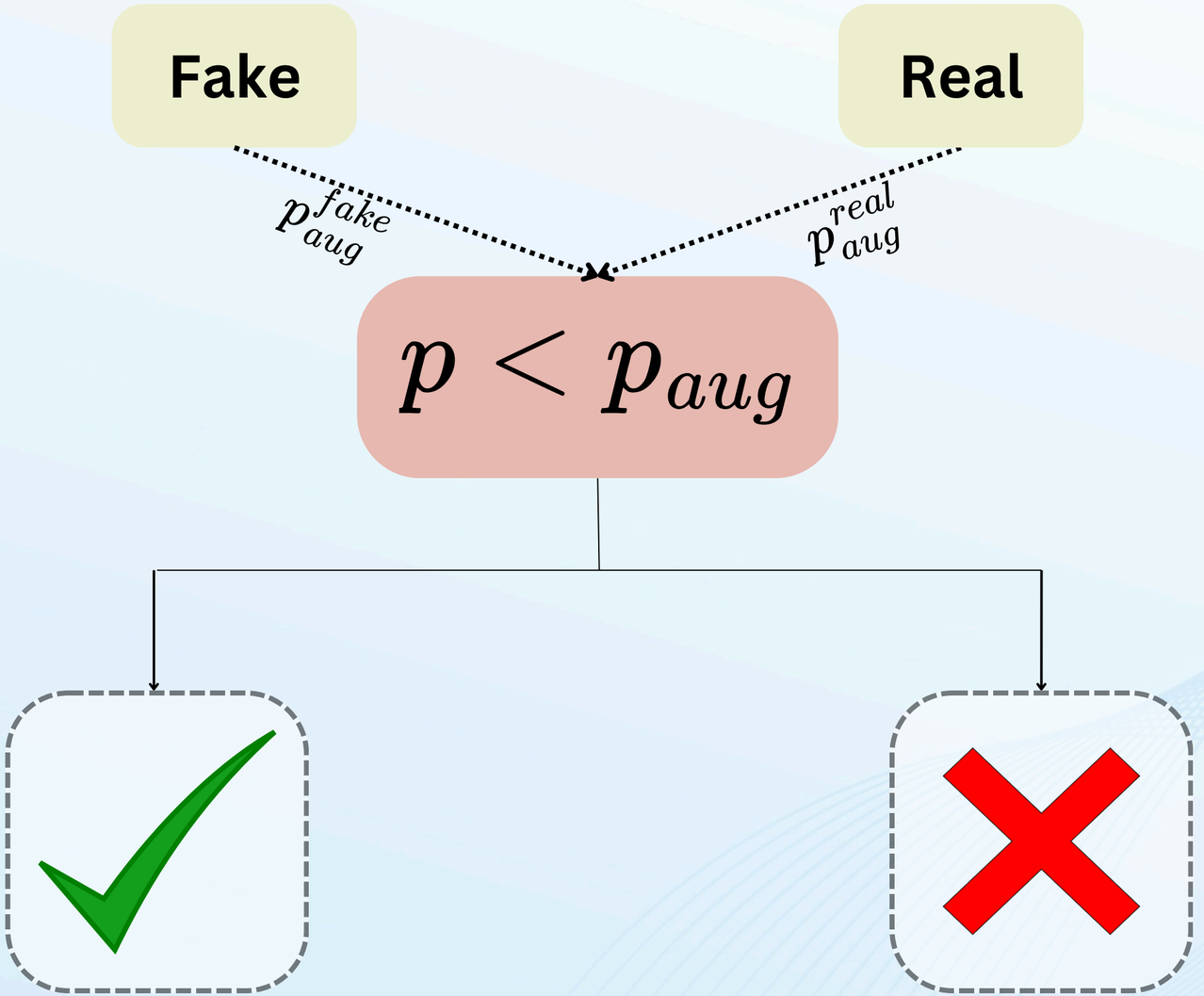
**Training tricks**

- Weighted Cross Entropy

**Optimizer**

- AdamW optimizer
- lr = 5e-5
- $\beta_1$= 0.9
- $\beta_2$= 0.999
- weight decay = 1e-4

# Augmentation



# Ablation Study

| Configuration | Accuracy % |
|---|---|
| Baseline | 94.36 |
| + weighted CE | 95.24 |
| + aug_prob ([0.3, 0.4]) | **96.97** |

+ 0.88 %

+ 1.73 %

Baseline + aug_prob ([0, 0.4]) + weighted CE : 91.92

Baseline + aug_prob ([0.3, 0.3]) + weighted CE : 94.95

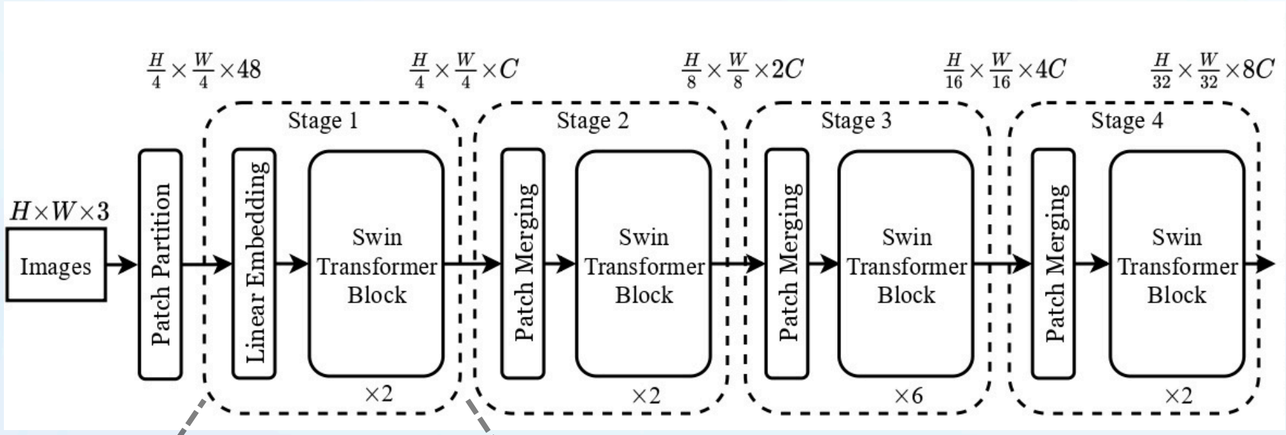
Baseline + aug_prob ([1, 1]) + weighted CE : 95.05

# Results

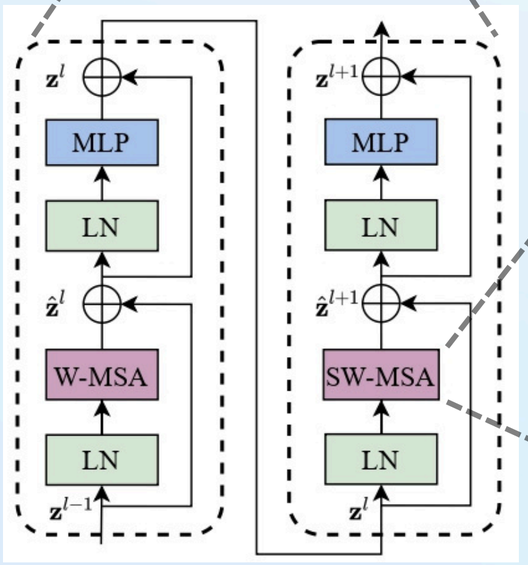
| Model | Accuracy % |
|---|---|
| ViT | 93 |
| ResNet-101 | 94.6 |
| EfficientNet-b4 | 91.08 |
| Inception-v4 | 94.63 |
| MaxViT | 96.09 |
| **ConvNeXt** | **96.97** |

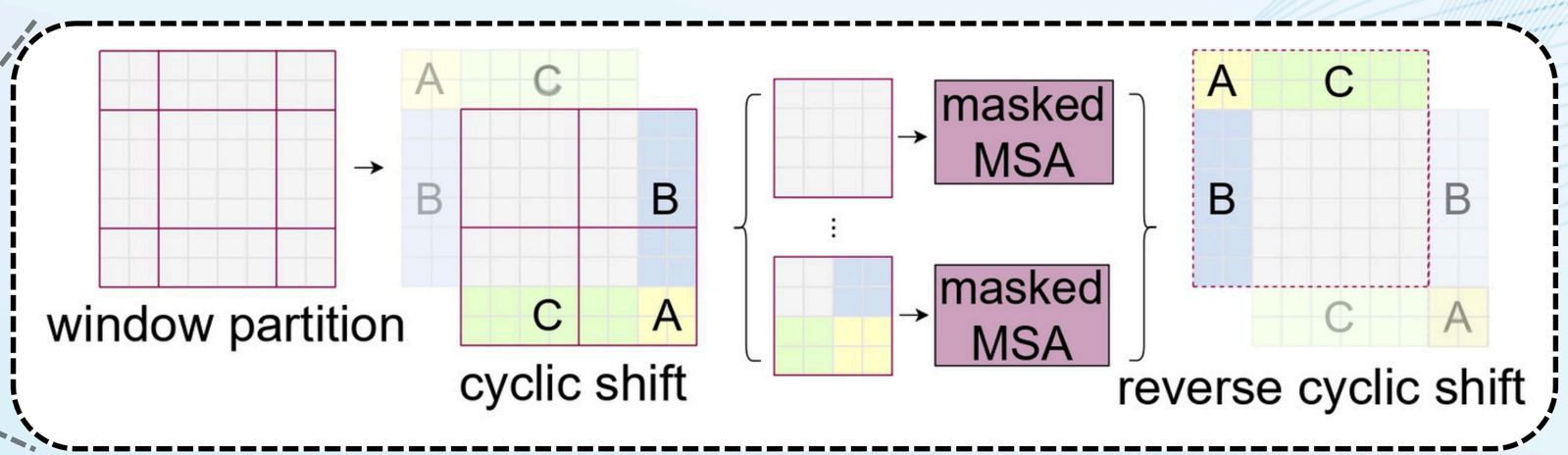
# Swin Transformer

# Swin Transformer




$$\text{Attention}(Q, K, V) = \text{SoftMax}\left(\frac{QK^T}{\sqrt{d}} + B\right)V$$




Liu, Z., Lin, Y., Cao, Y., et al. "Swin Transformer: Hierarchical Vision Transformer using Shifted Windows." ICCV, 2021

# Hyperparameters

**Basic**

- Batch size = 64
- Epochs = 10

**Augmentations**

- Rotation: [-10°, +10°]
- Gaussian Blur: 30% probability (Kernel: 3x5)
- Brightness & Contrast: 50% probability
- Horizontal Flip: 50% probability

**Training tricks**

- Weighted Cross Entropy

**Optimizer**

- AdamW optimizer
- lr = 5e-5
- $\beta_1$= 0.9
- $\beta_2$= 0.999
- weight decay = 1e-4

# Ablation Study

| Configuration | Accuracy % |
|---|---|
| Baseline | 93.71 |
| + weighted CE | 95.54 |
| + aug_prob ([0.3, 0.3]) | 95.63 |
| + aug_prob ([0.3, 0.5]) | **96.02** |

+ 1.83%

+ 0.09%

+ 0.39%

Baseline + weighted CE + aug_prob ([0.3, 0]) : 94.72

Baseline + weighted CE + aug_prob ([0, 0.3]) : 93.55

# Results

| Model | Accuracy % |
|---|---|
| ViT | 93 |
| ResNet-101 | 94.6 |
| EfficientNet-b4 | 91.08 |
| Inception-v4 | 94.63 |
| MaxViT | 96.09 |
| ConvNeXt | 96.97 |
| **Swin Transformer** | **96.02** |

# ENSEMBLE

# Ensemble Inference

- Used a **soft voting strategy** (weighted sum of confidence scores) to build ensembles of our best-performing models
- If the confidence scores output by the *k* models for an input **x** are $p_1, \ldots, p_k \in [0, 1]$ then the ensemble's inference is given by

$$f_{\text{ensemble}}(\mathbf{x}; \mathbf{w}) = \mathbf{1}\left\{\sum_{i=1}^{k} w_i p_i > 0.5\right\}$$

where $\mathbf{w} = (w_1, \ldots, w_k) \in [0, 1]^k$ are the ensemble weights with $\sum_{i=1}^{k} w_i = 1$

- $\sum_{i=1}^{k} w_i p_i$ is the ensemble's confidence score

# Finding Optimal Ensemble Weights

- Performed a grid search over combinations of $(w_1, \ldots, w_k)$ where each $w_i$ takes values in [0, 1] in intervals of 0.05

$$\begin{aligned} S &= \{0.05t \,|\, 0 \leq t \leq 20\} \\ &= \{0, 0.05, 0.10, \ldots, 0.95, 1\} \end{aligned}$$

- Picked the combination of ensemble weights that maximized the classification accuracy on the validation dataset $\{(\mathbf{x}_i, y_i)\}_{i=1}^{n}$

$$\mathbf{w}^* = \arg \max_{\substack{\mathbf{w} \in S^k \\ \sum_{j=1}^{k} w_j = 1}} \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{f_{\text{ensemble}}(\mathbf{x}_i; \mathbf{w}) = y_i\}$$

where $y_i$ is the ground truth label corresponding to $\mathbf{x}_i$

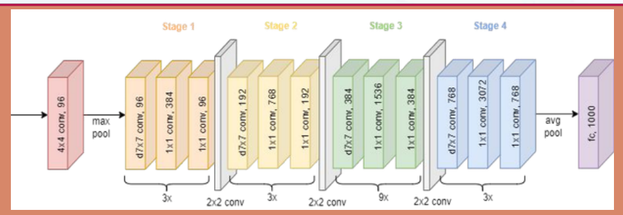
# Ensemble 1

Elon Musk

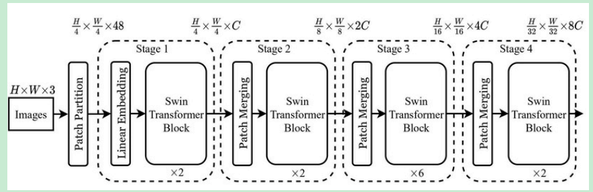
Real

Fake

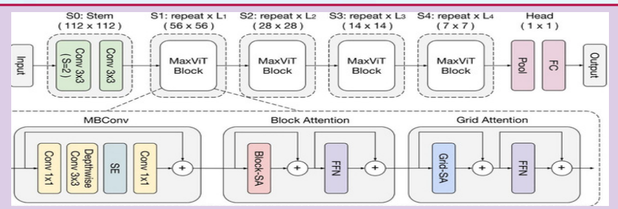
Elon Musk +
Nicolas Cage




ConvNeXt




Swin Transformer




MaxViT

Ensemble 1

# Ablation on Ensembles

| Ensemble | Accuracy (%) |
|---|---|
| MaxViT + Swin | 96.65 |
| MaxViT + ConvNeXt | 97.20 |
| Swin + ConvNeXt | 97.27 |
| MaxViT + Swin + ConvNeXt (class-based) | 97.23 |
| **Ensemble 1**: MaxViT + Swin + ConvNeXt (confidence-based) | **97.40** |

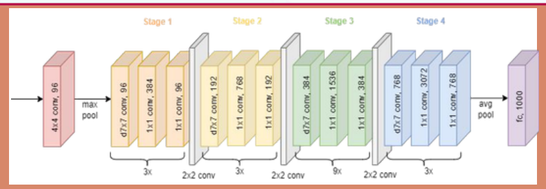
# Ensemble 2

Elon Musk

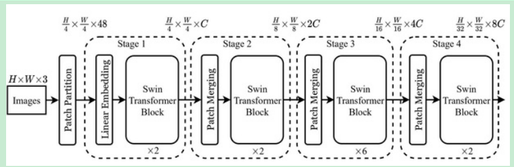
Real

Fake

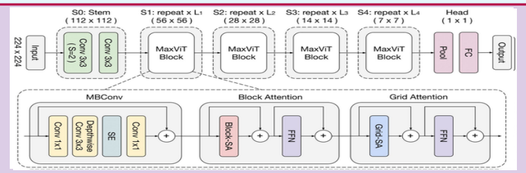
Elon Musk +
Nicolas Cage




ConvNeXt




Swin Transformer




MaxViT

OpenAI

GPT-4o

Ensemble 2

# Results

| Ensemble | #Parameters | Time* (ms) | Accuracy | AUC | F1-Score | Precision | Recall | EER |
|---|---|---|---|---|---|---|---|---|
| GPT-4o | - | - | 79.94 | 79.82 | 82.86 | 72.77 | 96.18 | 0.365 |
| Swin Transformer | 87.7M | 28.8 | 96.02 | 96.02 | 96.08 | 95.58 | 96.57 | 0.045 |
| MaxViT Tiny | 30.9M | 32.9 | 96.09 | 96.09 | 96.12 | 96.06 | 96.18 | 0.040 |
| ConvNeXt small | 50.2M | 7.22 | 96.97 | 96.95 | 97.05 | 95.32 | 98.83 | 0.049 |
| **Ensemble 1** | **168.8M** | **71.2** | **97.40** | **98.80** | **97.45** | **96.05** | **98.90** | **0.039** |
| Ensemble 2 | 168.8M+ | 71.2+ | 97.49 | 98.78 | 96.12 | 96.06 | 96.18 | 0.036 |

*Benchmarked on NVIDIA A5000

# EXPLAINABILITY

# Explainability

Real

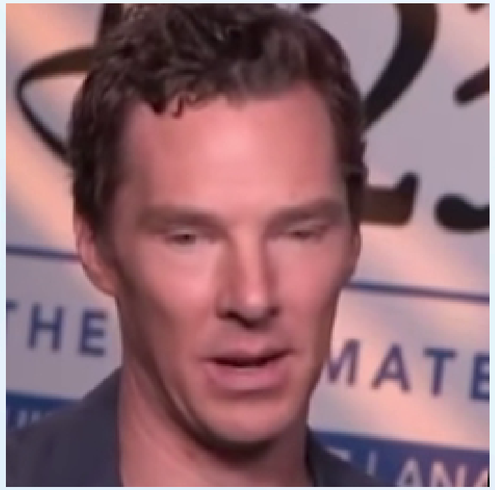
GradCAM

Fake

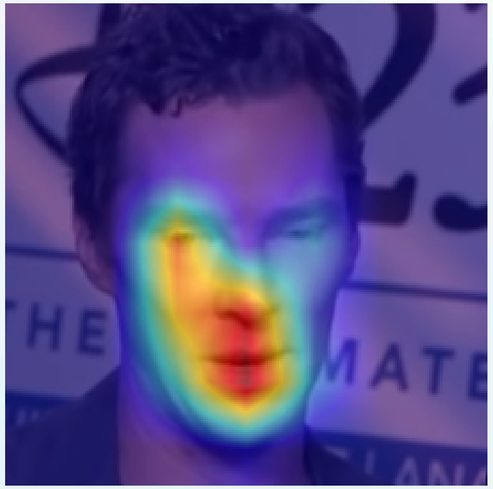
GradCAM

# Explainability

Real



GradCAM



Fake

GradCAM

# References


- Tu, Z., Talebi, H., Zhang, H., et al. "MaxViT: Multi-Axis Vision Transformer." ECCV, 2022
- Liu, Z., Mao, H., Wu, C. Y., Feichtenhofer, C., Darrell, T., & Xie, S. (2022). “A convnet for the 2020s.” CVPR, 2022
- Liu, Z., Lin, Y., Cao, Y., et al. "Swin Transformer: Hierarchical Vision Transformer using Shifted Windows." ICCV, 2021
- Yan, Z., Zhang, Y., Yuan, X., Lyu, S., & Wu, B. "DeepfakeBench: A Comprehensive Benchmark of Deepfake Detection." arXiv preprint arXiv:2307.01997, 2023.

# THANK YOU